Alessandro **Viespoli**

Student

(+39) 345 014 2480 | alessandro.viesp@gmail.com | zincalex | alessandroviespoli

Work Experience

Software Developer Engineer

Madrid, Spain

Amazon

06/2026 - Present

- Maintained a **ML pipeline** that detects emerging trends and matches them to relevant products across multiple international marketplaces.
- Contributed to an automated merchandising system that turns top-ranked trends into storefront content.
- Implemented a **model monitoring framework** for the production ML system that combines LLM-as-a-judge evaluation with human annotation, surfacing quality issues for action and building a labeled dataset for future model improvements.
- Applied **SDE best practices** to deliver high-quality, well-tested code through unit testing, code reviews and CI/CD pipelines.

Software Developer Engineer Intern

Madrid, Spain

Amazon

03/2025 - 09/2025

- Implemented **full-stack** solutions using Java, TypeScript, HTML/CSS, including comprehensive testing, monitoring, and documentation.
- Enhanced and optimized Amazon Business homepage widget leading to **61.5% increased user interactions** and **51.2% improvement in customer purchasing behavior**.
- Led technical investigation and **optimization of critical widget functionality**.
- Developed an AI-powered documentation assistant using Strands Agents SDK (Python) during an internal hackathon, creating a **multi-agent system** for standardizing wiki content creation through code base knowledge access; project selected for internal team deployment.

Education

University of Padua

Padua, Italy

MSc in Computer Engineering - AI & Robotics

10/2023 - 3/2026

- **Main courses**: Deep Learning, Computer Vision, Machine Learning, Artificial Intelligence, NLP, Reinforcement Learning, Intelligent robotics, Industrial robotics, Learning from networks, Big Data, Game theory, Operation Research, Automata Languages.
- Final score: 110/110 with honors (GPA 4.0)

University of Padua

Padua, Italy

BSc in Computer Engineering

09/2023

- **Main courses**: Artificial Intelligence, Databases, Data structures & algorithms, Operative Systems, Computer architectures, Computer networks, Control Theory, Electronics, Discrete Math, Probability, Linear Algebra, Calculus 1 & 2, Physics 1 & 2, Project Management, Software Engineering.
- Final score: 110/110 with honors (GPA 3.76)

Projects

Training-Free Depth Pruning of Vision Transformers

Python, PyTorch, Hugging Face

- Extended similarity-based layer dropping from LLMs to vision transformers (ViT, DINOv2, DINOv3, SwinV2). Removed 20% of DINOv2 blocks without retraining, giving 24% higher throughput at under 1% relative accuracy loss. [**Code**]

Parking Space Occupancy using Computer Vision

C++, OpenCV, CMake

- Build a program that automatically detect parking spaces, determine the occupancy and segment all cars within frame only using computer vision techniques. Achieved an average 0.75 mAP and 0.9 mIoU.[**Code**]

Land Cover Classification using Sentinel-2 Satellite Images

Python, Pytorch

- Developed different DL techniques to classify images based on 13 channels. Ensemble of ResNet-50 & Densenet-161 architectures and PCA analysis. Leading to 96.9% accuracy. [**Code**]

Named Entity Recognition System for general purpose tagging

Python, Pytorch

- Supervised Fine Tuned Gemini 2 model to identity and tag entities in phrases with different techniques through zero-shot or few-shot learning. Accuracy and recall of 91% in few-shot technique. [**Code**]

Autonomous Navigation and Manipulation with TIAGo

C++, ROS1

- Programmed the TIAGo robot to explore a lab, find requested AprilTags and report their positions on the map. It then picks objects with MoveIt and places them along a given line. [**Code**]

Achivements

Huawei Seeds for the Future Program *Remote*

10/2022

- Selected as one of best 50 skilled Italian ICT talents to participate in a learning program. Top 2 finalists of the team Hackathon ”Tech4Good”.

Skills

- **Programming Languages**: Python, C++, Java, C, HTML/CSS/Typescript, SQL.
- **Libraries**: OpenCV, Pytorch, Apache Spark, Scikit-Learn and Strands Agents SDK.
- **Development Tools & Environment**: AWS, GitHub, CMake, Linux.
- **Languages**: English level C1 (IELTS score 7.5 and Cambridge).